Chapter 35: The Influence of Monetary and Fiscal Policy on Aggregate Demand

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If the Fed dropped interest rates to zero tomorrow, do you think it would boost spending, and why might it fail?"

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Learning Objectives

35.1 Analyze economic shocks using the AD–AS and liquidity-preference models.

35.2 Explain how monetary policy affects the economy.

35.3 Explain how fiscal policy influences aggregate demand.

35.4 Evaluate the mechanisms and side effects of fiscal policy.

35.5 Determine appropriate stabilization policies for different economic conditions.

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Key Points

- **Interest Rates and Money Market:** Keynes's liquidity-preference theory explains that the interest rate adjusts to balance money supply and money demand.
- **Price Level and Aggregate Demand:** A higher price level increases money demand, raises interest rates, and reduces investment—producing the downward-sloping aggregate-demand (AD) curve.
- **Monetary Policy and AD:** Increasing the money supply lowers interest rates and shifts AD right; decreasing the money supply raises interest rates and shifts AD left.
- **Fiscal Policy and AD:** Higher government spending or lower taxes shift AD right, while lower spending or higher taxes shift AD left.
- **Magnitude of Fiscal Effects:** The multiplier effect amplifies fiscal policy's impact on AD, while crowding out offsets it by raising interest rates and reducing investment.
- **Stabilization Policy Debate:** Governments can stabilize the economy using monetary and fiscal tools, but economists disagree: supporters cite the need to counter AD shocks, while critics argue that policy lags can make stabilization destabilizing.

Introduction

- Earlier chapters covered:
  - the long-run effects of fiscal policy on interest rates, investment, economic growth
  - the long-run effects of monetary policy on the price level and inflation rate
- This chapter focuses on the short-run effects of fiscal and monetary policy, which work through aggregate demand.

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Aggregate Demand

- Recall, the *AD* curve slopes downward for three reasons:
  - The wealth effect
  - The interest-rate effect
  - The exchange-rate effect

the most important of these effects for the U.S. economy

- Next:
  A supply-demand model that helps explain the interest-rate effect and how monetary policy affects aggregate demand.

The Theory of Liquidity Preferences

- A simple theory of the interest rate (denoted ***r***).
- ***r*** adjusts to balance supply and demand for money.
  - This is different from the Classical Theory in which the price level adjusted to bring about a balance in the supply and demand for money
- **Money supply**: assume fixed by central bank, does not depend on interest rate.

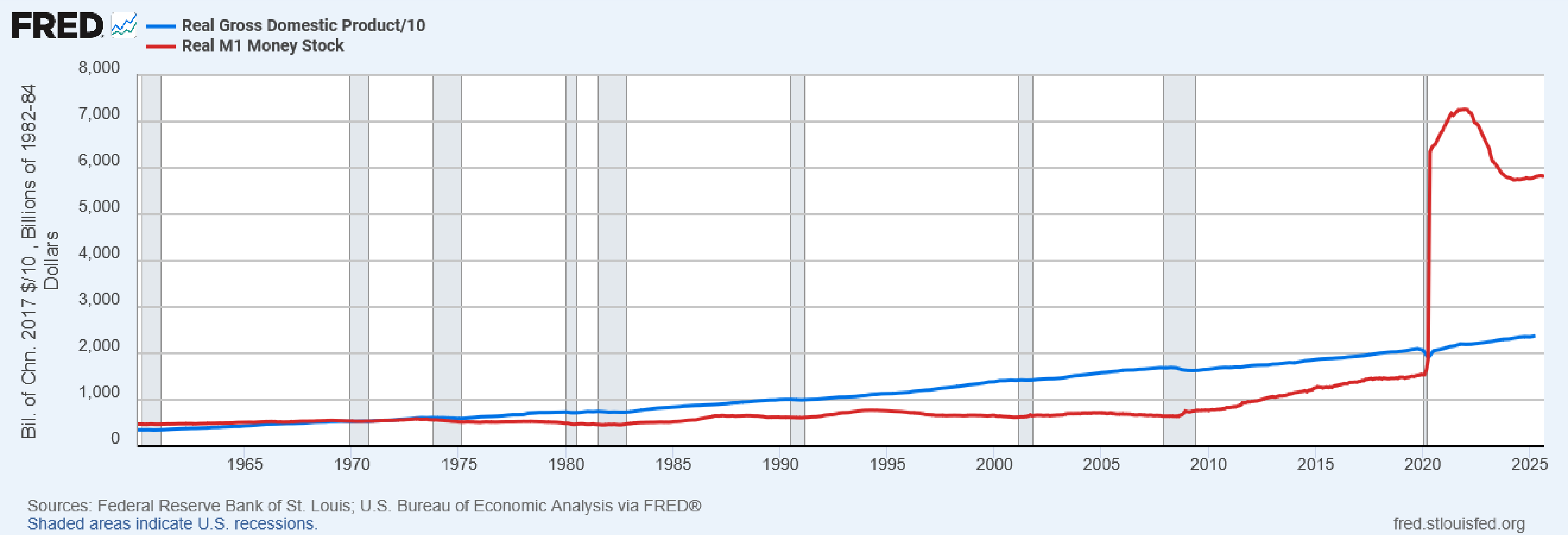
The Theory of Liquidity Preferences

- Money demand reflects how much wealth people want to hold in liquid form.
- For simplicity, suppose household wealth includes only two assets:
  - Money – liquid but pays no interest
  - Bonds – pay interest but not as liquid
- A household's "money demand" reflects its *preference* for *liquidity*.
- The variables that influence money demand: $\boldsymbol{Y}$, $\boldsymbol{r}$, and $\boldsymbol{P}$.

Money Demand

- Suppose real income (*Y*) rises. Other things equal, what happens to money demand?
- If *Y* rises:
  - Households want to buy more goods and services, so they need more money.
  - To get this money, they attempt to sell some of their bonds.
- *i.e.*, an increase in *Y* causes an increase in money demand, other things equal.

Income and Money Growth over Time



Activity: The Determinants of Money Demand: Classical Theory vs. Liquidity Preferences

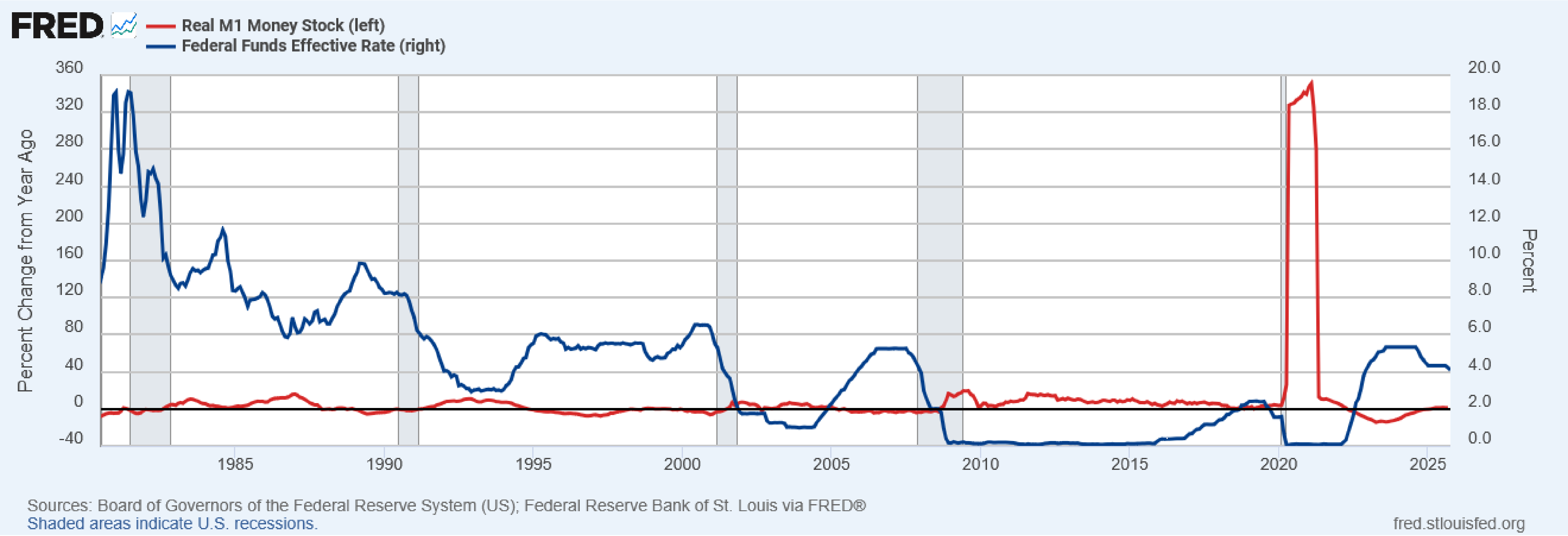
1. Suppose $\boldsymbol{r}$ rises, but $\boldsymbol{Y}$ and $\boldsymbol{P}$ are unchanged. What happens to money demand?
2. Suppose $\boldsymbol{P}$ rises, but $\boldsymbol{Y}$ and $\boldsymbol{r}$ are unchanged. What happens to money demand?

Activity: The Determinants of Money Demand: Classical Theory vs. Liquidity Preferences

1. Suppose r rises, but Y and P are unchanged. What happens to money demand?

- r is the opportunity cost of holding money.
  - An increase in r reduces money demand: households attempt to buy bonds to take advantage of the higher interest rate.
  - Had r fallen, money demand would have increased causing households to sell bonds
- Hence, an increase in r causes a decrease in money demand, other things equal.

Change in Money Stock and Fed Funds Rate



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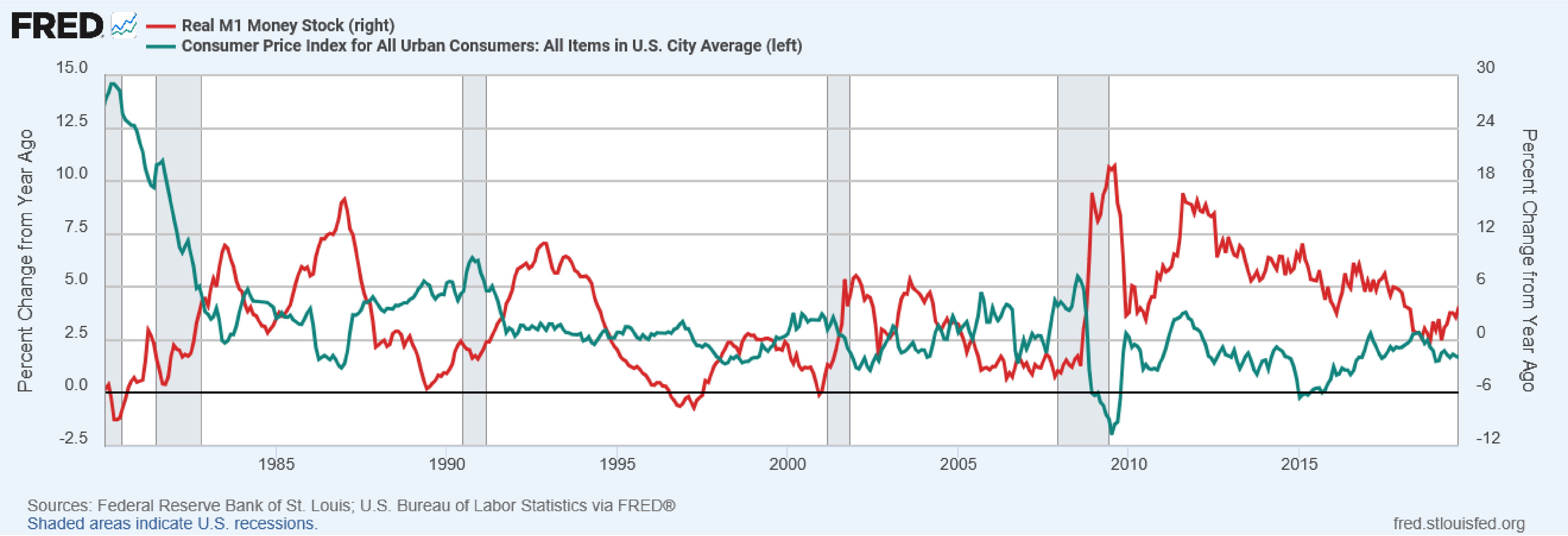
Activity: The Determinants of Money Demand: Classical Theory vs. Liquidity Preferences

2. Suppose ***P*** rises, but ***Y*** and ***r*** are unchanged. What happens to money demand?

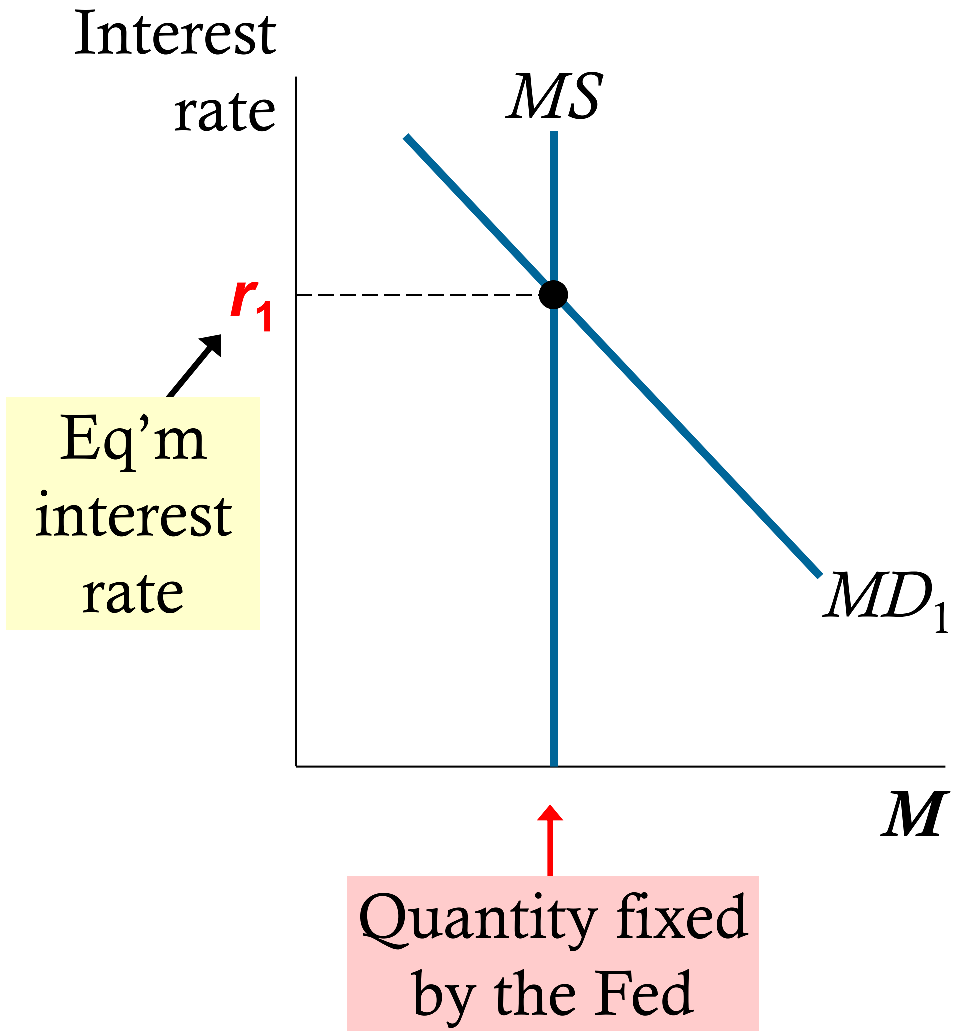
- If ***Y*** is unchanged, people will want to buy the same amount of goods and services.
  - Since ***P*** is higher, they will need more money to do so.
  - Had ***P*** fallen, people would need less money.
- Hence, an increase in ***P*** causes an increase in money demand, other things equal.

Inflation and Money Demand Growth



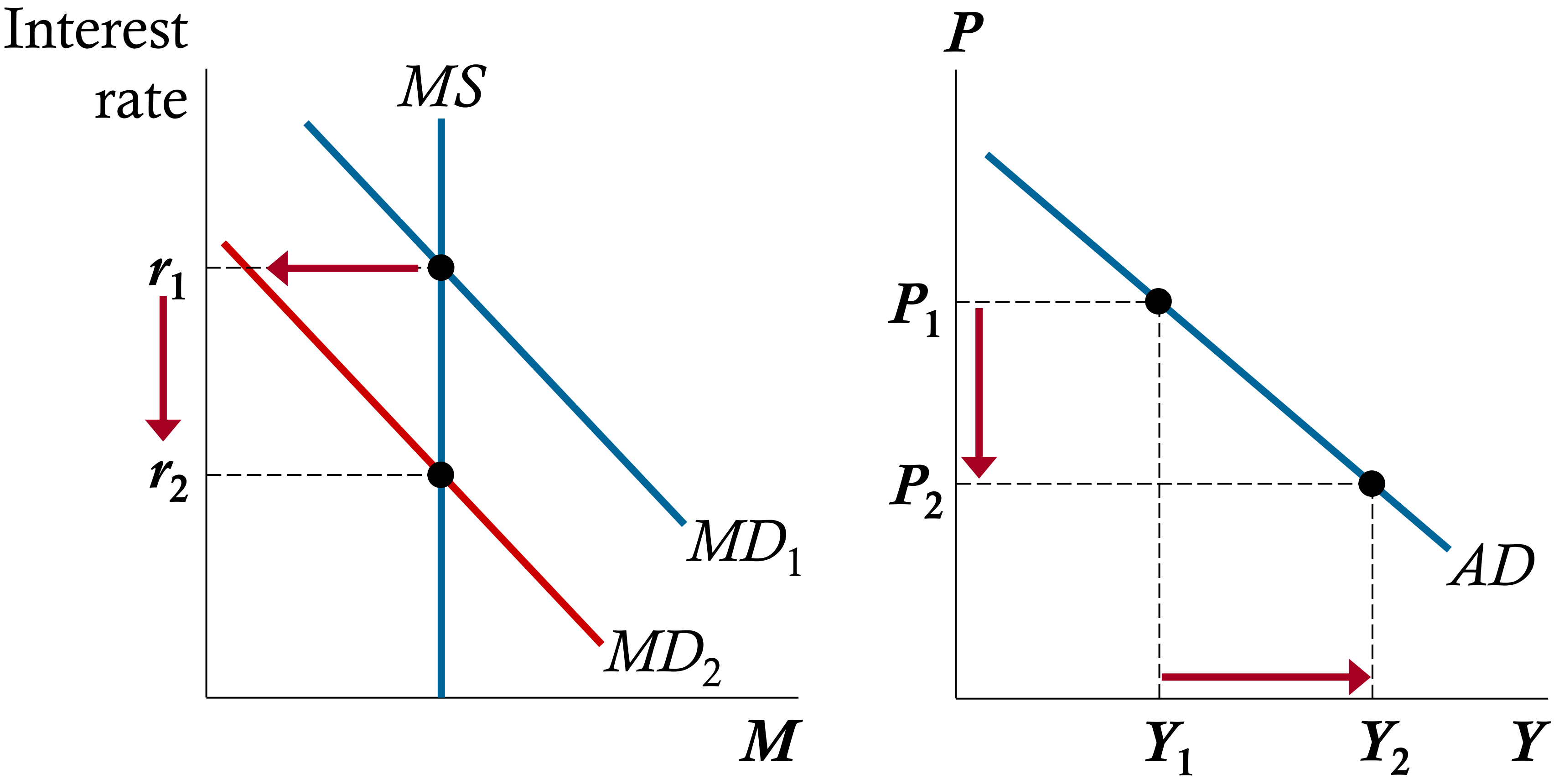
How is the interest rate (*r*) determined?

- *MS* curve is vertical: Changes in ***r*** do not affect *MS*, which is fixed by the Fed.
- *MD* curve is downward sloping: A fall in ***r*** increases money demand.



How the Interest Rate Effect Works

A fall in ***P*** reduces money demand, which lowers ***r***.



A fall in ***r*** increases ***I*** and the quantity of goods and services demanded.

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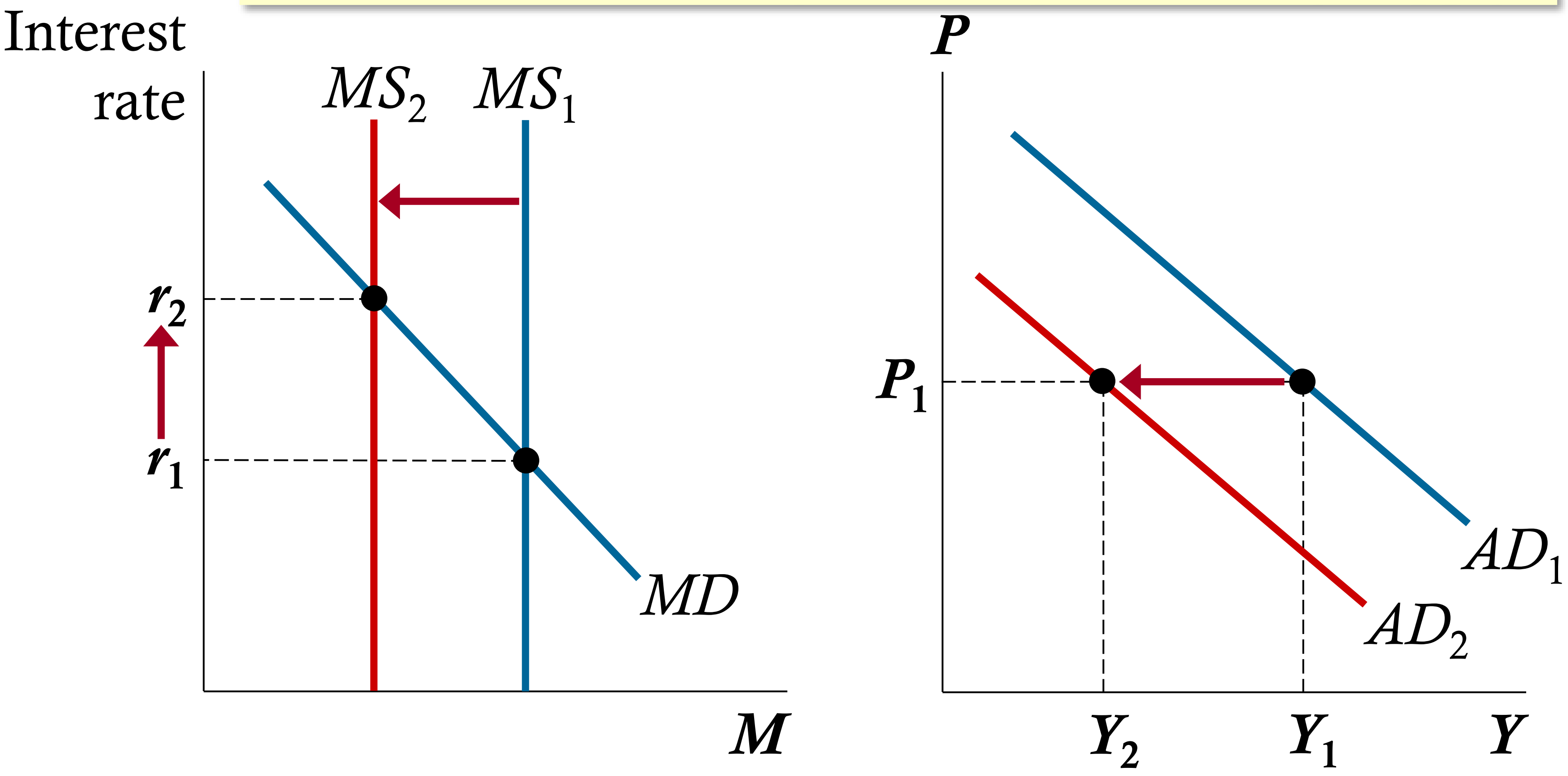
Monetary Policy and Aggregate Demand

- To achieve macroeconomic goals, the Fed can use monetary policy to shift the *AD* curve.
- The Fed's policy instrument is *MS*.
- The news often reports that the Fed targets the interest rate.
  - More precisely, the **federal funds rate**, which banks charge each other on short-term loans
- To change the interest rate and shift the *AD* curve, the Fed conducts open market operations to change *MS*.

The Effects of Reducing the Money Supply

The Fed can raise *r* by reducing the money supply.



An increase in *r* reduces the quantity of goods and services demanded.

Activity: Monetary Policy

For each of the events below,

- Determine the short-run effects on output
- Determine how the Fed should adjust the money supply and interest rates to stabilize output

1. Congress tries to balance the budget by cutting government spending.
2. A stock market boom increases household wealth.
3. A virus pandemic occurs causing business to shut down.

Activity: Monetary Policy

1. Congress tries to balance the budget by cutting government spending.

- This event would reduce aggregate demand and output.
- To stabilize output, the Fed should increase *MS* and reduce ***r*** to increase aggregate demand.

2. A stock market boom increases household wealth.

- This event would increase aggregate demand, raising output above its natural rate.
- To stabilize output, the Fed should reduce *MS* and increase ***r*** to reduce aggregate demand.

Answers

3. A virus pandemic occurs causing business to shut down.

- This event would reduce aggregate supply, causing output to fall.
- To stabilize output, the Fed should increase *MS* and reduce r to increase aggregate demand.

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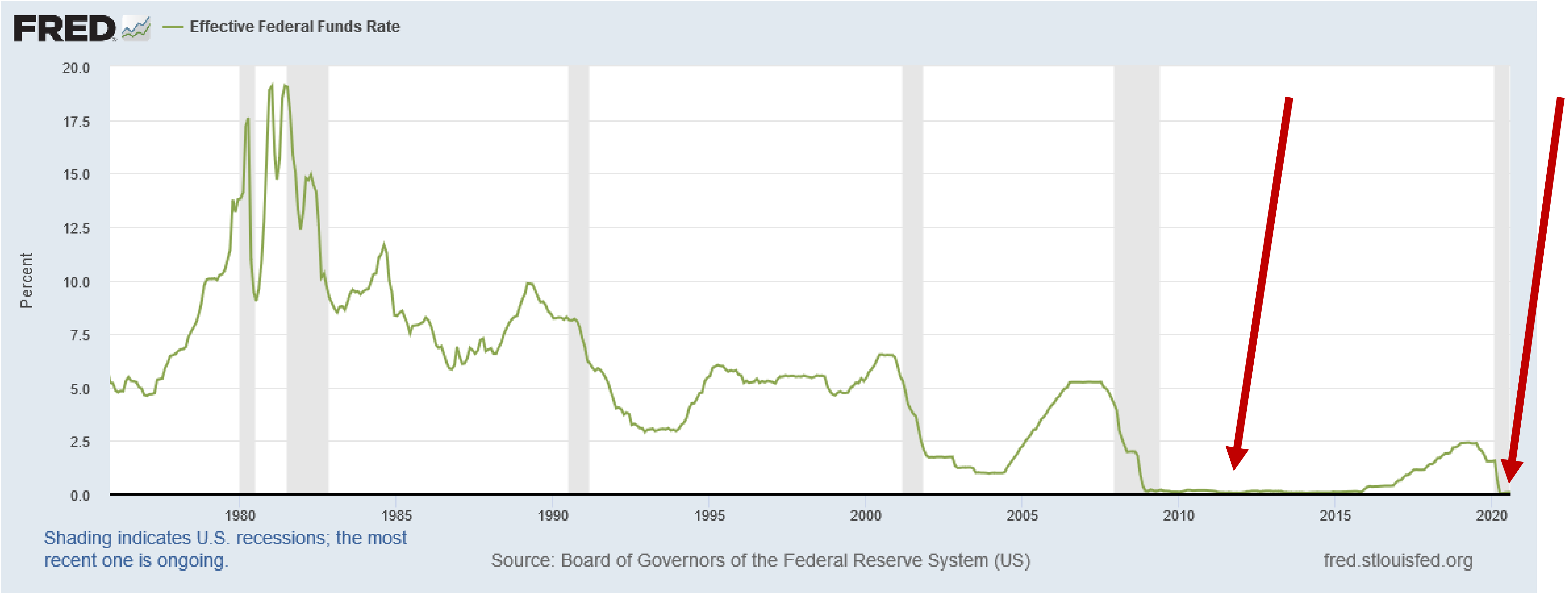
Liquidity Traps

- Monetary policy stimulates aggregate demand by reducing the interest rate.
- **Liquidity trap**: when the interest rate is zero
- In a liquidity trap, monetary policy may not work, since nominal interest rates cannot be reduced further.
- However, central bank can make real interest rates negative by raising inflation expectations.
- Also, central bank can conduct open-market operations using other assets—like mortgages and corporate debt—thereby lowering rates on these kinds of loans. The Fed pursued this option in 2008–2009 and 2020-2021.

Liquidity Trap



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Monetary Policy Summary

- Monetary policy affects aggregate demand by changing **interest rates**.
- Under **liquidity preference theory**, the interest rate adjusts to balance money supply and money demand.
- **Increasing the money supply** lowers interest rates, boosts investment, and shifts **AD right**.
- **Reducing the money supply** raises interest rates, lowers investment, and shifts **AD left**.
- The **interest-rate effect** explains why the AD curve slopes downward: higher prices → higher money demand → higher interest rates → lower spending.
- Policy effectiveness weakens near **zero interest rates**, creating a **liquidity trap** where traditional tools can’t stimulate demand.

Fiscal Policy and Aggregate Demand

- **Fiscal policy**: the setting of the level of government spending and taxation by government policymakers
- **Expansionary** fiscal policy
  - an increase in ***G*** and/or decrease in ***T***, shifts *AD* right
- **Contractionary** fiscal policy
  - a decrease in ***G*** and/or increase in ***T***, shifts *AD* left
- Fiscal policy has two effects on *AD*…

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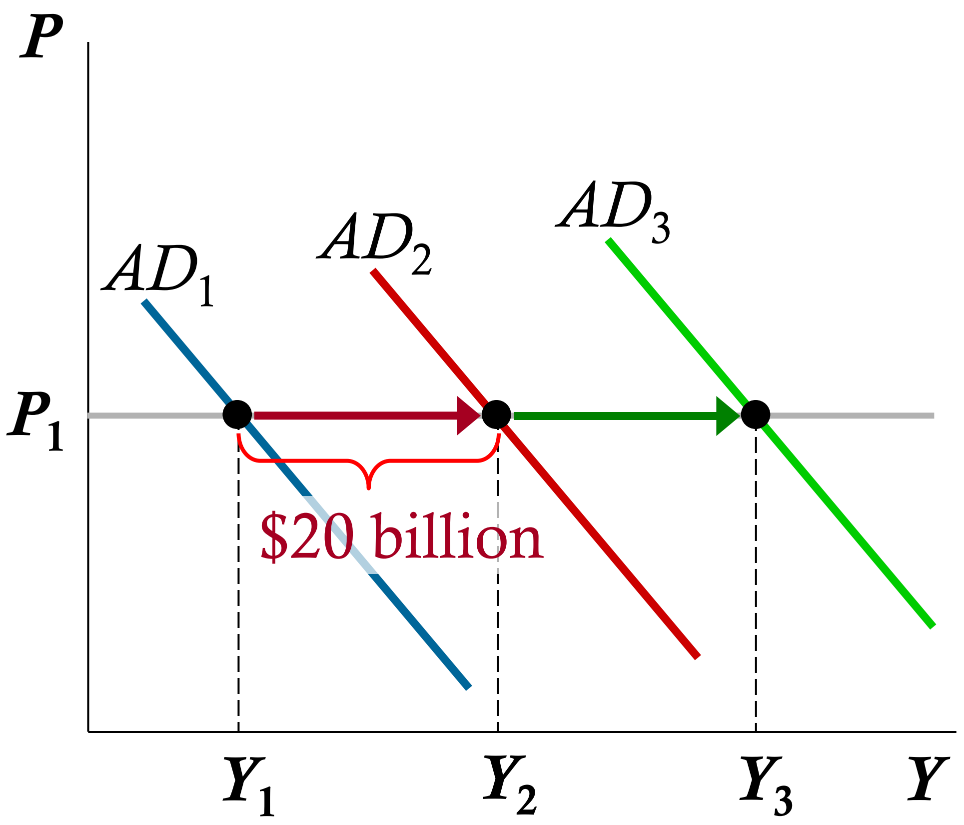
The Multiplier Effect

- If the government buys $20 billion of vaccines from Pfizer, Pfizer's revenue increases by $20 billion.
- This is distributed to Pfizer's workers (as wages) and owners (as profits or stock dividends).
- These people are also consumers and will spend a portion of the extra income.
- This extra consumption causes further increases in aggregate demand.

Multiplier effect: the additional shifts in *AD* that result when fiscal policy increases income and thereby increases consumer spending

The Multiplier Effect

- A \$20 billion increase in ***G*** initially shifts *AD* to the right by \$20 billion.
- The increase in ***Y*** causes ***C*** to rise, which shifts *AD* further to the right.



Marginal Propensity to Consume

- How big is the multiplier effect?
  - It depends on how much consumers respond to increases in income.
- **Marginal propensity to consume (MPC)**: the fraction of extra income that households consume rather than save
  - E.g., if $MPC = 0.8$ and income by rises \$100, $\boldsymbol{C}$ rises by \$80.
  - If $MPC = 0.5$ and income rises by \$100, $\boldsymbol{C}$ rises by \$50

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A Formula for the Multiplier

- Notation: $\Delta \boldsymbol{G}$ is the change in $\boldsymbol{G}$, $\Delta \boldsymbol{Y}$ and $\Delta \boldsymbol{C}$ are the ultimate changes in $\boldsymbol{Y}$ and $\boldsymbol{C}$
- $\boldsymbol{Y} = \boldsymbol{C} + \boldsymbol{I} + \boldsymbol{G} + \boldsymbol{NX} \rightarrow$ identity
- $\Delta \boldsymbol{Y} = \Delta \boldsymbol{C} + \Delta \boldsymbol{G} \rightarrow \boldsymbol{I}$ and $\boldsymbol{NX}$ do not change
- $\Delta \boldsymbol{Y} = MPC\, \Delta \boldsymbol{Y} + \Delta \boldsymbol{G} \rightarrow$ because $\Delta \boldsymbol{C} = MPC\, \Delta \boldsymbol{Y}$
- $\Delta \boldsymbol{Y} = \boxed{\dfrac{1}{1 - MPC}}\, \Delta \boldsymbol{G} \rightarrow$ solved for $\Delta \boldsymbol{Y}$

The multiplier

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A Formula for the Multiplier

- The size of the multiplier depends on *MPC*. For example,
  - if *MPC* = 0.5 $\rightarrow$ multiplier = 2
  - if *MPC* = 0.75 $\rightarrow$ multiplier = 4
  - if *MPC* = 0.9 $\rightarrow$ multiplier = 10

A bigger *MPC* means changes in ***Y*** cause bigger changes in ***C***, which in turn cause bigger changes in ***Y***.

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Other Applications of the Multiplier Effect

- The **multiplier effect**: Each \$1 increase in ***G*** can generate more than a \$1 increase in aggregate demand.
- Also true for the other components of GDP.
  - Example: Suppose a recession overseas reduces demand for U.S. net exports by \$10 billion.
  - Initially, aggregate demand falls by \$10 billion.
  - The fall in ***Y*** causes ***C*** to fall, which further reduces aggregate demand and income.

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Activity: Understanding the Multiplier

Scenario:

- Congress is considering a fiscal package to boost economic activity during a slowdown. Economists estimate the marginal propensity to consume (MPC) is **0.75**, and investment is expected to remain unchanged in the short run.

Questions:

1. Calculate the size of the spending multiplier if the MPC is 0.75.
2. Suppose Congress increases government purchases (G) by $50 billion. How much will aggregate demand ultimately increase?
3. If households suddenly become more cautious and reduce their MPC to 0.6, how does this change the size of the multiplier? What is the new total change in AD from the same $50 billion increase in G?

Activity: Understanding the Multiplier

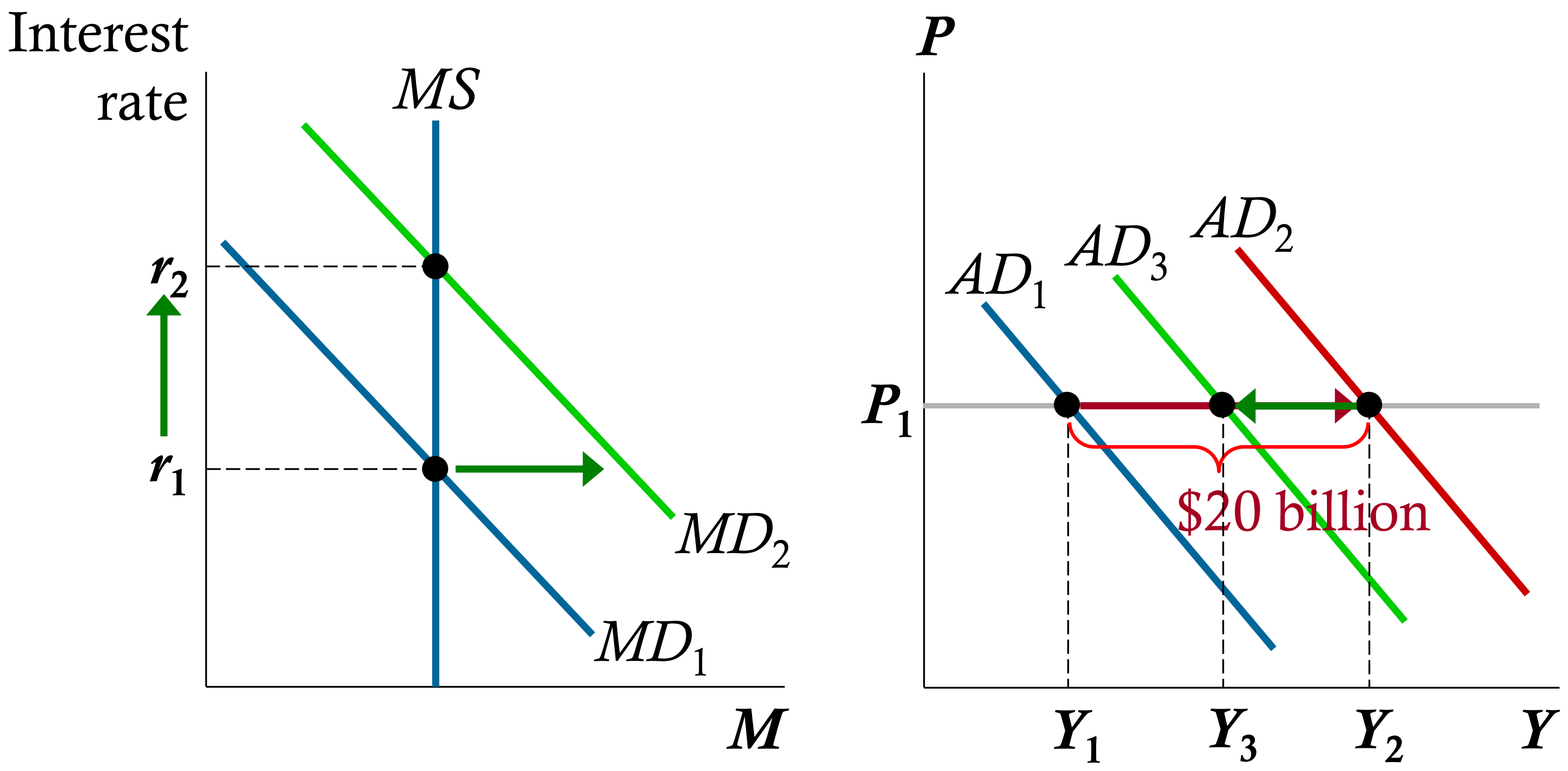
1. Multiplier = 1 / (1 – MPC) = 1 / (1 – 0.75) = 4
2. Total AD increase = \$50B × 4 = \$200B
3. New multiplier = 1 / (1 – 0.6) = 2.5
   Total AD increase = \$50B × 2.5 = \$125

The Crowding Out Effect

- Fiscal policy has another effect on *AD* that works in the opposite direction.
- A fiscal expansion raises ***r***, which reduces investment, which reduces the net increase in aggregate demand.
- So, the size of the *AD* shift may be smaller than the initial fiscal expansion.
- This is called the **crowding-out effect**.

How the Crowding Out Effect Works

A $20 billion increase in ***G*** initially shifts *AD* right by $20 billion



But higher ***Y*** increases *MD* and ***r***, which reduces *AD*.

Changes in Taxes

- A tax cut increases households' take-home pay.
- Households respond by spending a portion of this extra income, shifting *AD* to the right.
- The size of the shift is affected by the multiplier and crowding-out effects.
- Another factor: whether households perceive the tax cut to be temporary or permanent.
  - A permanent tax cut causes a bigger increase in ***C***—and a bigger shift in the *AD* curve—than a temporary tax cut.
  - The Tax Cuts and Jobs Act of 2018 is set to expire in 2025—if it doesn't get repealed.

Activity: Fiscal Policy Effects

Scenario:

The economy is in recession. Shifting the *AD* curve rightward by $200 billion would end the recession.

Questions:

1. If *MPC* = 0.8 and there is no crowding out, how much should Congress increase ***G*** to end the recession?
2. If there is crowding out, will Congress need to increase ***G*** more or less than this amount?

Activity: Fiscal Policy Effects

1. If *MPC* = .8 and there is no crowding out, how much should Congress increase ***G*** to end the recession?

- Multiplier = 1/(1 – .8) = 5
- Increase ***G*** by $40 billion to shift aggregate demand by 5 x $40 billion = $200 billion.

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Activity: Fiscal Policy Effects

2. If there is crowding out, will Congress need to increase ***G*** more or less than this amount?

- Crowding out reduces the impact of ***G*** on *AD*.
- To offset this, Congress should increase ***G*** by a <u>larger</u> amount.

Fiscal Policy and Aggregate Supply

- Most economists believe the short-run effects of fiscal policy mainly work through aggregate demand.
- But fiscal policy might also affect aggregate supply.
- Recall one of the Ten Principles from Chapter 1:
  - ***People respond to incentives.***
- A cut in the tax rate gives workers incentive to work more, so it might increase the quantity of goods and services supplied and shift *AS* to the right.
- People who believe this effect is large are called "Supply-siders."

Fiscal Policy and Aggregate Supply

- Government purchases might affect aggregate supply.
- Example:
  - government increases spending on roads.
  - Better roads may increase business productivity, which increases the quantity of goods and services supplied, shifts *AS* to the right.
- This effect is probably more relevant in the long run: it takes time to build the new roads and put them into use.

Using Policy to Stabilize the Economy

- Since the Employment Act of 1946, economic stabilization has been a goal of U.S. policy.
- Economists debate how active a role the government should take to stabilize the economy.

Fiscal Policy Summary

- Fiscal policy affects aggregate demand through **government spending** and **taxation**.
- **Expansionary fiscal policy** (↑G or ↓T) shifts **AD right**; **contractionary policy** (↓G or ↑T) shifts **AD left**.
- The **multiplier effect** amplifies changes in spending: new income → new consumption → larger total impact on AD.
- The **crowding-out effect** partially offsets fiscal expansion: higher income raises money demand → higher interest rates → reduced investment.
- Tax changes matter: **permanent tax cuts** shift AD more than temporary ones.
- Fiscal policy can influence **aggregate supply** over time through incentives (work, investment, productivity).

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The Case for Active Stabilization Policy

- Keynes: "Animal spirits" cause waves of pessimism and optimism among households and firms, leading to shifts in aggregate demand and fluctuations in output and employment.
- Also, other factors cause fluctuations, e.g.,
  - booms and recessions abroad
  - stock market booms and crashes
- If policymakers do nothing, these fluctuations are destabilizing to businesses, workers, consumers.

The Case for Active Stabilization Policy

- Proponents of active stabilization policy believe the government should use policy to reduce these fluctuations:
  - When GDP falls below its natural rate, use expansionary monetary or fiscal policy to prevent or reduce a recession.
  - When GDP rises above its natural rate, use contractionary policy to prevent or reduce an inflationary boom.

The Case against Active Stabilization Policy

- Monetary policy affects economy with a long lag:
  - Firms make investment plans in advance, so ***I*** takes time to respond to changes in ***r***.
  - Most economists believe it takes at least 6 months for monetary policy to affect output and employment.
- Fiscal policy also works with a long lag:
  - Changes in ***G*** and ***T*** require acts of Congress.
  - The legislative process can take months or years.

The Case against Active Stabilization Policy

- Due to these long lags, critics of active policy argue that such policies may destabilize the economy rather than help it:

  By the time the policies affect aggregate demand, the economy's condition may have changed.

- These critics contend that policymakers should focus on long-run goals like economic growth and low inflation.

Automatic Stabilizers

- **Automatic stabilizers**: changes in fiscal policy that stimulate aggregate demand when economy goes into recession, without policymakers having to take any deliberate action

Examples

1. The tax system
   - In a recession, taxes fall automatically, which stimulates aggregate demand.
2. Government spending
   - In recession, more people apply for public assistance (welfare, unemployment insurance).
   - government spending on these programs automatically rises, which stimulates aggregate demand.

Activity: Should the Government Stabilize the Economy?

Scenario:

- The economy has slipped into a mild recession after a decline in consumer confidence. Output is 1.5% below the natural rate, unemployment is rising, and tax revenues have begun falling automatically. At the same time, spending on unemployment insurance and income-support programs is increasing.
  - Automatic stabilizers have already kicked in, but should policymakers go further?

Questions:

1. Why might policymakers still pursue discretionary monetary or fiscal policy, even though automatic stabilizers are already responding?
2. How might policy lags (recognition, decision, implementation, and impact lags) make discretionary stabilization risky or counterproductive?
3. Identify two automatic stabilizers at work in this scenario and explain *how each* affects aggregate demand.

Activity: Should the Government Stabilize the Economy?

Suggested answers

1. Policymakers may still intervene because automatic stabilizers might not be strong enough to close the recessionary gap on their own. If output is falling and unemployment is rising, discretionary monetary or fiscal policy can provide an additional boost to aggregate demand. Active policy can help offset larger or faster-moving drops in spending, restore confidence, and prevent the recession from deepening. In short, stabilizers help, but they may not be sufficient in a significant downturn.
2. Discretionary policy can take too long to work. It may take months to recognize the downturn, longer for policymakers to debate and pass a response, and still longer for the policy's effects to be felt in the economy. By the time the policy takes effect, economic conditions may have changed possibly turning the policy into a source of instability. For example, stimulus passed after the economy is already recovering can push demand too high and contribute to inflation.

Activity: Should the Government Stabilize the Economy?

Suggested answers:
3. **Falling tax revenues:** When incomes decline during a recession, households and firms automatically pay fewer taxes. This leaves them with more disposable income, helping support consumption and slowing the decline in aggregate demand.

- **Increased government spending on unemployment benefits or social assistance:** As more people qualify for these programs, government spending rises automatically. This puts purchasing power directly into the hands of households most likely to spend it, supporting consumption and partially offsetting the reduction in aggregate demand.

Key Takeaways

- **Interest rates balance money supply and money demand.**
  - Liquidity preference theory explains how interest rates adjust in the short run.
- **The interest-rate effect drives the downward slope of aggregate demand.**
  - Higher prices increase money demand, raise interest rates, and reduce investment.
- **Monetary policy shifts the AD curve by changing interest rates.**
  - Increasing the money supply lowers interest rates and shifts AD right; decreasing it does the reverse.
- **Fiscal policy affects AD through spending, taxes, the multiplier, and crowding out.**
  - Government actions can amplify (multiplier) or dampen (crowding out) total spending.
- **Stabilization policy is powerful, and controversial.**
  - Economists debate whether government should actively offset economic fluctuations given long policy lags and uncertainty.

Knowledge Check

1. According to liquidity-preference theory, what causes interest rates to change?
2. Why does the aggregate-demand curve slope downward?
3. If the Fed increases the money supply, what happens in the short run?
4. What determines the size of the fiscal multiplier?
5. What is the crowding-out effect?
6. Why might stabilization policy fail to stabilize the economy?

Knowledge Check Answers

1. Interest rates adjust to equate money demand with money supply.
2. Wealth, interest-rate, and exchange-rate effects explain the downward slope.
3. More money → lower interest rates → higher investment → AD shifts right.
4. The MPC determines how income changes translate into spending changes.
5. Higher interest rates from government borrowing reduce private investment.
6. Long and variable lags in monetary and fiscal policy can destabilize the economy.